

SPACECRAFT ATTITUDE DYNAMICS AND CONTROL

Lab 4 – Attitude Kinematics

PROF. FRANCO BERNELLI
PIETRO BRUSCHI, MASSIMILIANO BUSSOLINO, ANNACHIARA IPPOLITO, ANDREA COLAGROSSI

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First Tutoring

Thursday October 15th
Room BL27.1.5
Hour: 14.15 -16.15

Write questions to me within this afternoon:
annachiara.ippolito@polimi.it

These two hours are meant to fill the gaps on the fundamentals of **MATLAB and Simulink**.
Partecipate if you have questions, it will not be a lecture but an interactive session!



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www.tutorapp.polimi.it -> try doing a first access!
Tomorrow you must register within the first half an hour.

Task 1: Simulate kinematics using the DCM with standard integration

Principal moments of Inertia

$I_x = 0.0700\ \mathrm{kg\,m^2}$
$I_y = 0.0550\ \mathrm{kg\,m^2}$
$I_z = 0.0250\ \mathrm{kg\,m^2}$

Initial conditions

$\omega_x(0) = 0.45\ \mathrm{rad/s}$
$\omega_y(0) = 0.52\ \mathrm{rad/s}$
$\omega_z(0) = 0.55\ \mathrm{rad/s}$

$$A(0) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Governing Equations

Dynamics

$$\begin{cases} \dot{\omega}_x = \dfrac{I_y - I_z}{I_x}\,\omega_y\omega_z \\ \dot{\omega}_y = \dfrac{I_z - I_x}{I_y}\,\omega_x\omega_z \\ \dot{\omega}_z = \dfrac{I_x - I_y}{I_z}\,\omega_y\omega_x \end{cases}$$

Kinematics

$$\frac{\mathrm{dA}(t)}{\mathrm{d}t} = -[\omega]^{\wedge}\mathrm{A}(t)$$

With

$$[\omega]^{\wedge} = \begin{bmatrix} 0 & -\omega_z & \omega_y \\ \omega_z & 0 & -\omega_x \\ -\omega_y & \omega_x & 0 \end{bmatrix}$$

Tasks

- Simulate the kinematics starting trying different values of the inertia tensor and initial conditions
- Test the orthonormality of the propagated solution

Task 2: Orthonormalize the matrix at each timestep

Traditional numerical integration procedures introduce numerical errors that affect the orthonormality of the solution. A possible solution is to use **structure-preserving integrators.** Another solution is to orthonormalize the solution at each timestep:

$$\mathrm{A}_{k+1}(t) = \frac{3}{2}\mathrm{A}_k(t) - \frac{1}{2}\mathrm{A}_k(t)\mathrm{A}_k^{\mathrm{T}}(t)\mathrm{A}_k(t)$$

This approach converges rapidly, with increasing k, to the exact value of A.
In a first order approximation it is possible to adopt a single step iteration:

$$\mathrm{A}(t) = \frac{3}{2}\mathrm{A}_0(t) - \frac{1}{2}\mathrm{A}_0(t)\mathrm{A}_0^{\mathrm{T}}(t)\mathrm{A}_0(t)$$

Tasks

- Test the orthonormality of the propagated solution after orthonormalization

Task 3: Simulate kinematics using quaternions

Principal moments of Inertia

$I_x = 0.0700\ \text{kg m}^2$

$I_y = 0.0550\ \text{kg m}^2$

$I_z = 0.0250\ \text{kg m}^2$

Initial conditions

$\omega_x(0) = 0.45\ \text{rad/s}$

$\omega_y(0) = 0.52\ \text{rad/s}$

$\omega_z(0) = 0.55\ \text{rad/s}$

$\boldsymbol{q}(0) = [0, 0, 0, 1]^{\text{T}}$ (Scalar last)

Governing Equations

Dynamics

$$\begin{cases} \dot{\omega}_x = \dfrac{I_y - I_z}{I_x}\ \omega_y \omega_z \\ \dot{\omega}_y = \dfrac{I_z - I_x}{I_y}\ \omega_x \omega_z \\ \dot{\omega}_z = \dfrac{I_x - I_y}{I_z}\ \omega_y \omega_x \end{cases}$$

Kinematics

$$\dot{\boldsymbol{q}}(t) = \frac{1}{2}\Omega(t)\boldsymbol{q}(t)$$

with

$$\Omega(\text{t}) = \begin{bmatrix} 0 & \omega_z & -\omega_y & \omega_x \\ -\omega_z & 0 & \omega_x & \omega_y \\ \omega_y & -\omega_x & 0 & \omega_z \\ -\omega_x & -\omega_y & -\omega_z & 0 \end{bmatrix}$$

Tasks

- Simulate the kinematics starting trying different values of the inertia tensor and initial conditions
- Test the normality of the propagated solution